8 Features of manufacturing industries

8.3 Modern manufacturing methods and systems when designing for manufacture for small, medium and large-scale production

Topic	What students need to learn:
8 Features of manufacturing industries	8.3 – Characteristics, processes, application, advantages and disadvantages and the importance of considering accuracy of production and efficiency of modern manufacturing methods and systems when designing for manufacture for small, medium and large scale production: *a) production scheduling and production logistics* *b) robotics in production – robots on fully-automated production and assembly lines/cells*

Topic	What students need to learn:
8 Features of manufacturing industries	*c) materials handling systems – automated storage and retrieval systems (ASRS), automatic guided vehicles (AGVs)* *d) flexible manufacturing systems (FMS), modular/cell production systems* *e) lean manufacturing using just-in-time (JIT) systems* *f) standardised parts, bought-in components* *g) quick response manufacturing (QRM)* *h) data integration – product data management (PDM), enterprise resource planning (ERP) systems* *i) concurrent manufacturing.*

a) Production Scheduling and Production Logistics

Production scheduling and production logistics

Production, planning and control (PPC) – computer systems are used to:

- *Plan and control production*
- *Organise the availability of materials and components*
- *Coordinate suppliers and customers*

Production scheduling and production logistics

Logistics

- **Logistics** is the often complex organisation of the flow of all the elements involved in manufacturing and marketing products, from parts and materials to the product reaching the customer.
- An efficient ***supply chain network (SCN)*** is vital to ensure the flow of materials and components required to manufacture products on a global scale and particularly to operate a **Just in Time (JIT) system**.

Production scheduling and production logistics

Master Production Schedule (MPS)

- **Master Production Schedule *(MPS)*** software carries out several co-ordinated PPC functions in a JiT environment including the following:

Availability of Parts and Materials

- *Triggers and coordinates orders to suppliers of materials and controls stock inventory*
- *JiT production requires suppliers of materials and components to deliver stock only when needed.*

Availability of Parts and Materials Cont…

- To facilitate JiT system the flow of materials and components through a factory must be carefully controlled using a **'Kanban'** (signal card) system for reordering components. (originally a Toyota initiative)
- Many of the ***Kanban cards*** that were once used have now been replaced by ***barcode labels, radio frequency identification (RFID) tags and scanners linked to* Master Production Schedule *(MPS) software***.
- In the car industry, parts are often delivered to where they are needed by overhead gantries and conveyors at the precise time they are needed.

Availability of Parts and Materials Cont…

- As cars are assembled, their progress is tracked using a telemetry device such as RFID linked to the Master Production Schedule (MPS) computer.

Scheduling of Machines and People

- Planning production schedules and timings networking with departments such as personnel to ensure the appropriate availability of suitably skilled labour.
- Automatic reconfiguration of manufacturing plants to facilitate ***QRM*** *(Quick Response Manufacturing)* and the re-programming of ***FMS cells*** *(Flexible Manufacturing System)* to facilitate, e.g. switching between different car models to suit JiT orders.

Co-Ordinating Suppliers and Customers

- Links to electronic point of sale (EPOS) for automatic transmission of JiT manufacturing requirements.

Advantages: Production Scheduling and Production Logistics
▪ *Flexible and easily adaptable if product mix or quality changes* ▪ *Minimise work in progress and reduce inventory* ▪ *Maintain balance on production line* ▪ *Raise productivity levels*

Video: Line Conveyor…



Now Test Yourself:

1. List ***four*** ways that ***Master Production Schedule (MPS)*** software helps to ***coordinate just-in-time (JiT) manufacture***.